σ^*

The Resolution Calibration Principle: Certified Locality for Kernel and Gibbs Fields

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Abstract

A prediction that superposes stored points through distance-decaying weights depends, in principle, on every point. We show it can be made certifiably *local*: past a chosen radius, the influence of all farther sources is provably below a budget ε, with no labels. The certificate is one measured inequality, read at three levels. For an additive kernel field it is closed-form, $\sigma^*=d/\sqrt{2\ln(A/\varepsilon)}$, with the interfering mass *A* measured from the field and conservative from one measurement; the same bound certifies decision stability, safe deletion and abstention. For normalised Gibbs fields--softmax attention and prototypical networks--a companion temperature certificate bounds the far probability mass directly, and a label-free conformal wrapper extends the guarantee to the next query. On frozen vision, text and tabular fields the certificate holds where label-free selectors miss the budget by orders of magnitude, and it re-certifies without labels as a memory grows. Bandwidth selection becomes calibration.

A predictor that reads a space through distance-decaying weights over stored points depends, in principle, on all of them: every stored point contributes to every query. We build such predictors expecting them to be *local*--to answer from the near points and leave the rest undisturbed--but nothing in the construction guarantees it. This paper supplies that guarantee, in closed form and without labels: past a chosen radius, the total influence of all farther sources is provably below a budget ε. Certified locality is the advance; the bandwidth is only the dial that delivers it. The guarantee is one measured inequality, and it connects a single geometric quantity to controlled data dependence, stable decisions, safe deletion, abstention, and--as a memory grows--label-free re-certification.

The object is the *kernel field*, one of the most reused constructions in applied mathematics. Place weighted sources at points z_i of a metric space, attach to each a kernel of common width σ, and read the field at a query *y* as the superposition of the nearby kernels. Kernel density estimators, Nadaraya–Watson regressors, radial-basis interpolants (whose shape parameter is the same bandwidth choice), Gaussian-process posteriors, and the local correction fields used to adapt frozen models are all instances of this one object, sharing one free parameter--the bandwidth σ, the resolution at which the field reads the space. Too fine and it responds only at its sources; too coarse and it bleeds across the space and washes out the structure it was meant to resolve. That resolution is chosen, almost always, by search: a grid, a random sample, or a cross-validated sweep, each scored by instantiating the field and measuring an outcome--a held-out likelihood, a validation loss, a labelled accuracy--and none returning a guarantee about the value it selects.

We show the search is unnecessary. The question it answers--``which σ scores best on held-out labels''--is not the one locality requires; the operative question is ``which σ keeps the field's cross-source interference below a stated budget'', and that has a closed-form, label-free answer. The *certified frontier* $\sigma^*=d/\sqrt{2\ln(A/\varepsilon)}$ is the largest bandwidth a conservative aggregate envelope certifies to hold the interference past a radius *d* below ε, with the effective mass *A measured* from the field, not tuned. Its structural core is that this measured mass is nondecreasing in σ, so a single measurement is conservative: one pass, no labels, and a closed-form a-priori guarantee. The exact tail admits a slightly larger bandwidth (our own label-free search finds ≈*1.1*×), so σ^* trades a little bandwidth for the closed form and the proof. The same inequality reads four ways at once--safety, decision stability, abstention, and data *locality*, the last certifying that the prediction depends only on the training points within *d*.

The certificate uses only that the output superposes distance-decaying weights, so it is not confined to density estimation. For *normalised* readouts--softmax attention, whose weight on a key is an inner product rather than a distance, and prototypical networks, which are kernel fields already--a companion certificate bounds the far probability mass directly in the logit gap, carrying the same calibration onto pretrained transformers and few-shot learners. On three frozen vision backbones and four further modalities the additive certificate holds the budget where the label-free selectors we tested miss it, and matches a labelled grid search using zero labels. And because the mass is re-measured rather than tuned, the certificate re-certifies a field it has never labelled or tuned on: as a fixed memory densifies, σ^* tracks a per-stage re-tuned labelled oracle with zero labels while a bandwidth frozen by a one-time search drifts to hundreds of times the budget--the adaptivity a one-time labelled search structurally lacks in the regime of continually-updated models and growing memories.

RESULTS

THE CERTIFICATE

Kernel fields.

Fix a metric space, taken here as R^n. A *kernel field* is a finite collection $F=(z_i,v_i)_{i=1}^M$ of sources $z_i\in R^n$ carrying signed weights $v_i\in R$, together with a Gaussian kernel of common bandwidth σ, written in squared distance as

$$K_\sigma(t) = \exp(-t/2\sigma^2), t = \|y - z_i\|^2$$

The field's value at a query *y* is the superposition $\Sigma_i\ v_i\ K_\sigma(\|y\text{-}z_i\|^2)$. The base certificate controls this *unnormalised* sum; the normalised readouts of Section 1.3 (softmax attention, prototype and Nadaraya–Watson fields) are covered separately by the Gibbs certificate proved there, which bounds normalised far mass directly. Density estimators, interpolants, and the correction fields of Section 1.2 are all fields of this unnormalised form. Write $V_{max}=max_i|v_i|$; the certified quantity is invariant under $(v_i,\varepsilon)(cv_i,c\varepsilon)$, so we fix $V_{max}=1$ (the budget ε is in units of the largest source weight), which holds exactly for the one-hot Parzen vote of Section 1.2.

Near and far, and the effective mass.

Around a query y, a separation radius d partitions the sources: those within d are *near* (the field should respond to them), those in $F=i{:}\|\ y-z_i\|>d$ are *far* (it should leave them undisturbed). The far tail $Tail_\sigma(y,d)=\Sigma_{i\in F}|v_i|\ K_\sigma(\|\ y-z_i\|^2)$ is the field mass leaking past d; the bandwidth question is how large σ can be before it exceeds a budget ε. Not every far source matters equally: most are negligibly weak, and a handful near the radius carry the tail. The right count is therefore not $|F|$ but the number of far sources that *effectively* contribute--the participation ratio of the far weights $\omega_i=K_\sigma(\|\ y-z_i\|^2)$, which equals the true count when they are equal and drops toward one when a few dominate,

$$N_{\mathrm{eff}}(\sigma)=\frac{(\sum_{i\in\mathcal{F}}\omega_i)^2}{\sum_{i\in\mathcal{F}}\omega_i^2}\in[1,|\mathcal{F}|]$$

which must be restricted to F: a participation ratio over all sources is dominated by the near mass and does not bound the tail. The field mass acting past the radius is $A=V_{max}N_{eff}\geq 1$.

Assumptions.

The certificate rests on three explicit conditions. (A1) *Bounded weights*: $|v_i|\leq V_{max}=1$. (A2) *A meaningful near/far radius*: $d>0$ partitions sources into a near set the field should respond to and a far set it should not, so every far source satisfies $\omega_i\leq K_\sigma(d^2)$. (A3) *Budget below mass*: $0<\varepsilon<A$. It is worth separating three kinds of statement in what follows: the frontier and the safety, stability, and conservativeness results are *proven* under (A1)–(A3); the effective mass A and the tail ratios are *measured* from the field; and the radius d (here the tenth percentile of pairwise centre distances) is a *design choice*--any fixed rule gives a valid certificate. One further scope note on (A1): the one-shot conservativeness (Proposition 2) assumes the source amplitudes v_i are fixed as σ varies. For radial-basis interpolants and Gaussian-process posteriors whose fitted coefficients change with the length scale, the certificate is therefore directly valid as a *post-hoc* guarantee at a fixed fitted scale; using it for bandwidth *selection* on such systems requires an envelope on how the coefficients grow, which we do not develop here.

The safety frontier.

Define the leakage certificate $cert(A,\sigma,d)=A\ K_\sigma(d^2)=A\ e^{-d^2/2\sigma^2}$.

Proposition 1 (Safety frontier). *For $A>0$, $d>0$, $0<\varepsilon<A$, the equation $cert(A,\sigma,d)=\varepsilon$ has the unique solution*

$$\sigma^*(A,\varepsilon,d)=\frac{d}{\sqrt{2\ln(A/\varepsilon)}}$$

and the safe set $\sigma>0{:}cert\leq\varepsilon$ is $(0,\sigma^*]$.

Sketch. cert is continuous and strictly increasing in σ, rising from 0 to A; solving $cert=\varepsilon$ gives the unique frontier, and monotonicity makes the safe region $(0,\sigma^*]$ (SI Note 1). The single-source bound extends to the whole tail: because every far source lies beyond d, each far kernel is bounded by $K_\sigma(d^2)$, and the participation-ratio mass converts this per-source bound into an aggregate one,

$$\mathrm{Tail}_\sigma(y,d)=\sum_{i\in\mathcal{F}}|v_i|\omega_i\leq V_{\max}N_{\mathrm{eff}}K_\sigma(d^2)=\mathrm{cert}(A,\sigma,d)$$

with $A=V_{max}N_{eff}$; at $\sigma=\sigma^*$ the aggregate leakage is at most ε (SI Note 1).

The measured mass is self-consistent.

The frontier depends on $A=V_{max}N_{eff}(\sigma)$, but N_{eff} itself depends on σ, so A must be measured at some σ and then used to set σ^*. We measure it once at the pairwise scale $\sigma_{pair}=d/\sqrt{(2\ln(1/\varepsilon))}$ (the frontier for a single far source) and deploy at σ^*. This one-shot procedure is provably conservative:

Proposition 2 (Conservativeness). *$N_{eff}(\sigma)$ is nondecreasing in σ. Consequently $\sigma^*\leq\sigma_{pair}$, the mass measured at σ_{pair} satisfies $A(\sigma_{pair})\geq A(\sigma^*)$, and the true tail at the deployed bandwidth obeys $Tail_{\sigma^*}(y,d)\leq A(\sigma_{pair})\ K_{\sigma^*}(d^2)=\varepsilon$.*

Sketch. As σ grows the far weights become more uniform, so their participation ratio cannot decrease; since $\sigma_{pair}\geq\sigma^*$, measuring the mass there overestimates the deployed mass, and the one-shot certificate holds (SI Note 2). On the three frozen backbones the one-shot mass over-estimates the deployed mass by *2.2–4.3*× (Fig. 1a) and the measured one-shot tail is *0.08–0.16* ε--strictly inside budget on every network. The exact self-consistent bandwidth, when wanted, is the unique root of the monotone envelope $A(\sigma)K_\sigma(d^2)=\varepsilon$ and is found by bisection; the naive re-measurement iteration settles empirically in six to seven steps (Fig. 1b), but the one-shot value already certifies (SI Note 2). Both load-bearing facts--the participation-ratio tail bound and this monotonicity--are confirmed independently on random configurations (zero violations in *20,000* and *3,000* trials respectively; SI Note 8).

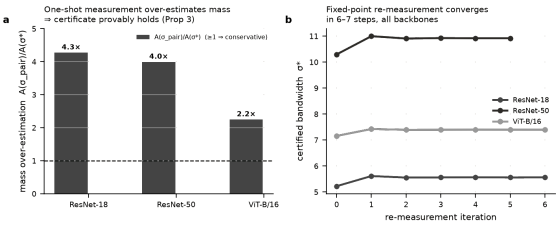

Figure 1. The measured mass is conservative. (a) Measuring A once at σ_{pair} over-estimates the mass at σ^* by *2.2–4.3*× on all three backbones, so the one-shot certificate provably holds (Proposition 2). (b) The re-measurement sequence settles in *6–7* steps; the exact self-consistent bandwidth is the root of the monotone envelope, found by bisection.

The same bound controls the decision.

Consider a field used as a classifier: each class c receives a score $S_c(y)=\Sigma_{i:\ class(i)=c}\ v_i\ K_\sigma(\|\ y-z_i\|^2)$, with prediction $_c\ S_c$. Splitting each score by the near/far partition, the certificate bounds the far part of every class score by ε at $\sigma=\sigma^*$. Let the *local prediction* be $_c\ S_c^{near}$ and the *local margin $m(y)$* the gap between the top two near-scores.

Theorem 1 (Decision stability). *If $\sigma\leq\sigma^*$ and $m(y)>2\varepsilon$, the deployed prediction equals the local prediction: the far field cannot change the decision.*

Sketch. Each class score moves by at most ε under far-field interference, so a two-class gap moves by at most 2ε; a near margin above 2ε cannot be flipped (SI Note 3). This is the perturbation bound underlying interval-based certified robustness ; what is new is *where* ε *comes from*--a bandwidth calibrated so the far-field perturbation is provably at most ε, read from geometry without labels. The test uses only near scores and ε, so a deployer flags certified queries per query, and a floor follows:

Corollary 1 (Performance floor). *At $\sigma\leq\sigma^*$, deployed accuracy ≥ (near-field accuracy) - [$m(y)\leq 2\varepsilon$].*

The same inequality certifies *data locality*. Let $F'(y)$ be the field with its far set F arbitrarily deleted, or replaced by any *admissible* corrupted far set: sources of bounded weight $|v_i'|\leq V_{max}$ located outside the radius d whose effective mass satisfies $N_{eff}'\leq A$. (Corrupting the weights of the far sources in place leaves their positions, hence N_{eff}, unchanged and is always admissible; the condition only excludes an adversary who repositions far mass to inflate N_{eff}' past the measured budget.)

Corollary 2 (Far-set locality). *At $\sigma\le\sigma^*$: (i) deleting the far set moves the output by at most the budget, $\|F(y)-S(y)\|=\|T(y)\|\le\varepsilon$; (ii) replacing it by any admissible far set shifts each class score by at most ε, so $\|F'(y)-F(y)\|\le 2\varepsilon$; and (iii) if $m(y)>4\varepsilon$ the decision is unchanged under any such deletion or corruption. The prediction is therefore ε-insensitive to the presence, and 2ε-insensitive to the identity, of every training point beyond d.*

Sketch. (i) is Proposition 1 verbatim: $Tail_\sigma(y,d)\le\varepsilon$ *is* $\|T(y)\|\le\varepsilon$. For (ii) each admissible far set has $N_{eff}'\le A$, so it contributes at most $V_{max} N_{eff}' K_\sigma(d^2)\le V_{max} A\, K_\sigma(d^2)=\varepsilon$ to any class score by the same certificate (the admissibility bound $N_{eff}'\le A$ is exactly what keeps the corrupted tail inside the budget); the original and corrupted far contributions each lie in a ball of radius ε and differ by at most 2ε; (iii) is then Theorem 1 applied to a 2ε perturbation. All three hold empirically on four datasets, and fail by two to three orders of magnitude at a bandwidth *3*× too wide (Fig. 2).

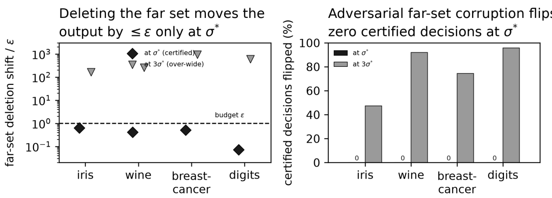

Figure 2. Certified data locality (Corollary 2). (a) At σ^* deleting the entire far set moves the output by $\le\varepsilon$ on all four datasets; a bandwidth *3*× too wide overshoots by two to three orders of magnitude. (b) Under worst-case adversarial corruption of the far set, zero certified-stable decisions flip at σ^*, versus *47–96%* at $3\sigma^*$.

One certificate, four readouts.

At $\sigma=\sigma^*$ every query splits the field as $F(y)=S(y)+T(y)$ with $\|T(y)\|\le\varepsilon$--which *is* Proposition 1. Because $S(y)$ is precisely the response the field would give with the entire far set deleted, the same inequality reads four ways:

- *Safety*: the deployed response is within ε of the near-field response, $\|F(y)-S(y)\|\le\varepsilon$.
- *Performance*: if the near margin exceeds 2ε, $F=S$ (Theorem 1).
- *Abstention*: if the top near score is $\le 2\varepsilon$ the certificate cannot fire, and the query lies outside the field's certified support--a label-free per-query support signal.
- *Locality*: the output is ε-insensitive to the presence, removal, or corruption of *every* far source--deleting the entire far set moves $F(y)$ by at most ε, and no admissible far set can flip a decision with near margin above 4ε (Corollary 2). This is a measured, per-configuration data-locality guarantee: it certifies *which* training points a prediction can depend on, and connects the bandwidth to unlearning and attribution rather than to density estimation. It is not a differential-privacy bound--there is no randomisation and the guarantee is over the field as measured, not worst-case over datasets--but it occupies the same role for distance-based predictors.

We confirmed the decomposition on *20,000* certified-regime configurations with zero violations, and on a real frozen ResNet-18 field (*20,000* kernels, *10,000* queries) the certified σ^* flags *27%* of queries stable with a stable-implies-equal rate of *100.00%*--zero violations--while the naive bandwidth certifies nothing and its accuracy has collapsed. The certified-stable set sits on the field's accuracy–coverage curve rather than above it; its distinguishing property is that the selection rule is label-free and a priori (SI Note 6). A Lipschitz argument extends the pointwise bound to a ball: the certified radius is genuine but small ($\delta\approx$*1.6–2.2%* of inter-query distance), so the per-query certificate carries the deployment claims (SI Note 4).

THE CERTIFICATE IN PRACTICE

The certificate was built for the dense regime, where many sources act at once and the naive single-interferer intuition fails. We reproduce that failure, and its cure, on real frozen deep networks; the field construction, the three bandwidths compared, and the label-free computation of σ^* are specified in Methods.

Results.

The pattern is identical across all three architectures (Fig. 3, Table 1). The naive bandwidth overshoots the interference budget by three to four orders of magnitude (tail/ε from 4.9×10^3 to 1.0×10^4) and collapses the field into a single component spanning 97–100% of the centres. The certified σ^* holds the budget (tail/ε between *0.08* and *0.16*), dissolves the giant component to 4–7%, and recovers accuracy to within 1–5 points of the labelled grid-search optimum--using *zero labels*, with ε fixed across every backbone so the adaptation to *512*, *768*, or *2048* dimensions happens automatically through the measured mass *A*. The classical selectors are established methods, not straw comparators, and they miss this budget: Silverman and Scott over-smooth heavily (tail/ε$>10^5$, accuracy 27–64%, *below* the naive rule), Abramson's rule more so (12–36%). The two label-free distance heuristics--the median heuristic and a *k*-nearest-neighbour scale, the natural competitors to σ^*--pick a bandwidth *3.6–6*× larger, which drives tail/ε to 10^5 and loses *9–26* accuracy points: the quantile *d* fixes the *scale*, but the logarithmic factor $\sqrt{2\ln(A/\varepsilon)}$ is what turns it into a *safe* one, and only the calibration supplies it. Over five re-samples the certified accuracy varies by *≤0.3* points on every backbone (*56.3±0.2*, *51.1±0.2*, *72.2±0.2 %*), so the ordering is not seed-luck.

The certificate is robust to its own design choices.

Four checks, all label-free, confirm the operating point is not fragile (SI Note 5). *From average to per-query.* The mass *A* is a query-average, so the certificate is stated on the average tail; in deployment one cares about the *worst single query*. Empirically the two coincide: over *10,000* ResNet-18 queries the largest single-query tail/ε is *0.24*, still well inside budget, so no query exceeds it. To make the per-query guarantee a priori rather than empirical, set *A* to an upper quantile of N_{eff} instead of its mean; this costs *≤0.2* points of accuracy and gives zero exceedances on all three backbones. The near set is *25–47*× class-enriched (weighted near purity *0.31/0.25/0.47* against a chance level of *0.01*), though this is a *plurality*: the certificate guarantees *which* sources decide, not that they decide correctly, and accuracy remains a separately measured outcome. And accuracy stays flat across the ε×percentile grid (*≤6.7* points, over a realized σ^* range of only *1.34–1.41*×; SI Note 5) while the certified tail stays in budget--a conjunction that a merely accuracy-tuned dial would not show, since a bandwidth *4–6*× larger leaves the budget by orders of magnitude.

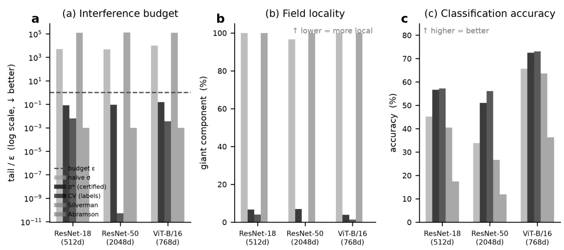

Figure 3. Certified σ^* across three frozen backbones (CIFAR-100, ε=*0.05*, zero labels). (a) Only σ^* holds the interference budget (tail/ε<*1*); naive and classical KDE bandwidths overshoot by 10^3–10^5×. (b) Naive and KDE bandwidths collapse the field into one component; σ^* keeps it local. (c) σ^* matches the labelled grid-search accuracy label-free, while

the KDE selectors trail badly.

Backbone	Method	tail/ε	giant%	acc%
ResNet-18	naive	5068	100	45.2
(512d)	median heur.	215695	100	39.6
	k-NN(7)	120953	100	40.5
	σ*	0.08	7	56.6
	grid (labels)	0.01	4	57.2
	Silverman	125520	100	40.4
ResNet-50	naive	4892	97	33.9
(2048d)	median heur.	210717	100	25.3
	k-NN(7)	129605	100	26.6
	σ*	0.09	7	51.0
	grid (labels)	0.00	0	56.1
	Silverman	127973	100	26.6
ViT-B/16	naive	9963	100	65.7
(768d)	median heur.	216464	100	63.1
	k-NN(7)	110309	100	63.7
	σ*	0.16	4	72.5
	grid (labels)	0.00	1	73.0
	Silverman	123721	100	63.6

Table 1. Certified σ^* versus naive, median-heuristic, *k*-nearest-neighbour, labelled grid-search, and classical density bandwidths on three frozen backbones. The two label-free distance heuristics--the natural competitors to σ^*--pick a bandwidth *3.6–6*× too large, overshooting the interference budget by roughly five orders of magnitude and losing *9–26* accuracy points; σ^* is the only label-free rule that holds tail ≤ε and stays near the labelled grid-search accuracy.

Beyond vision: four non-image modalities.

The certificate is a property of the kernel-field geometry, not of the image domain. We test it on four corpora unrelated to CIFAR (Table 2, Fig. 4), each over five seeds: three text collections--DBpedia, AG-News, Yahoo-Answers--embedded with a frozen sentence transformer (*384*-d), and a raw tabular set (Covertype, *54*-d). The naive bandwidth overshoots by 3.6×10^2 to 2.3×10^3 while σ^* holds tail/ε<*1*, and every stable-flagged query has its deployed prediction equal to the intended one (*100%* on every corpus). On the three high-dimensional text sets σ^* matches the labelled grid search to within *0.9* points *label-free*; on the low-dimensional tabular set it trails by *5.8* points, the same specificity-first behaviour as the public datasets. The pattern the flagship establishes on vision reproduces, with error bars, on text and tabular data alike.

Corpus (dim)	method	tail/ε	acc%
DBpedia-14	σ*	0.24	92.1,±0.6
text (384)	grid (lab.)	---	92.3,±0.5
	median	45609	89.0,±0.7
	k-NN(7)	33492	89.1,±0.6
AG-News-4	σ*	0.27	87.9,±1.1
text (384)	grid (lab.)	---	88.0,±1.1
	median	13018	85.7,±0.8
	k-NN(7)	10272	85.8,±0.9
Yahoo-10	σ*	0.30	66.9,±0.5
text (384)	grid (lab.)	---	67.8,±0.3
	median	32631	65.8,±0.4
	k-NN(7)	24822	66.0,±0.4
Covertype-7	σ*	0.10	65.0,±0.8
tab. (54)	grid (lab.)	---	70.8,±1.1
	median	20755	53.9,±1.5
	k-NN(7)	102	60.0,±1.6

Table 2. The certificate generalises across modality (5 seeds, ε=*0.05*; naive omitted for space, tail/ε 3.6×10^2–2.3×10^3). σ^* holds tail/ε<*1* label-free and matches the labelled grid search within *0.9* points in high dimension; the *5.8*-point trail on low-dimensional tabular data is the same specificity-first regime as Section 1.4. The label-free median and *k*-NN heuristics overshoot the budget by 10^2–10^5× on every corpus (the same failure as Table 1), with an accuracy cost of *1–3* points on text and up to *11* on tabular. The stable-implies-equal rate is *100%* throughout.

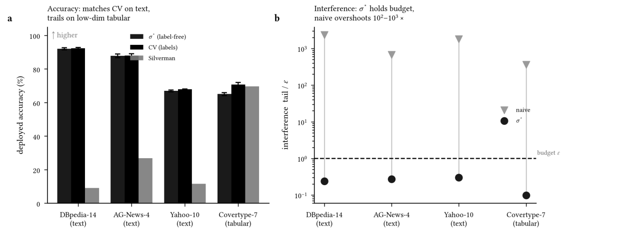

Figure 4. Across four non-image modalities. (a) σ^* (label-free) matches the labelled grid search on the three text sets and trails only on low-dimensional tabular data; error bars are ±*1* s.d. over five seeds. (b) σ^* holds the interference budget on every corpus, while the naive bandwidth overshoots by two to three orders of magnitude.

Beyond the Gaussian: monotone, log-concave tails.

The certificate uses the Gaussian shape only lightly, and the two halves need different amounts. The frontier itself--inverting $A K_\sigma(d)=\varepsilon$--needs only that the far weight $\omega_i=K_\sigma(\|y-z_i\|)$ be monotone and invertible in σ. One-shot conservativeness needs more: the normalised far-weight vector must flatten in majorization order as σ grows, which holds exactly when *ln K* is concave as a function of *ln* distance (SI Note 2). So any invertible monotone tail admits a frontier; the log-distance-concave kernels additionally admit conservative one-shot mass measurement. The Laplace kernel $K_\sigma(r)=e^{-r/\sigma}$ is one such: its frontier is $\sigma^*=d/\ln(A/\varepsilon)$--the same radius-over-a-log-factor structure, with the square-root gone because the tail is exponential in *r* rather than r^2--and its far weights still flatten as σ grows, so N_{eff} is nondecreasing and one measurement still over-estimates. We verify this on iris, wine, breast-cancer, and digits (Fig. 5): the Laplace σ^* holds tail/ε<*1* on all four (mean *0.16–0.24*), matches a labelled grid search anchored to its own scale within *0–2.8* points, and shows zero monotonicity violations. The principle is a property of log-concave-tail kernel fields, not of the Gaussian shape alone.

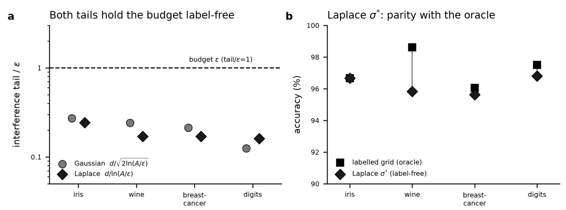

Figure 5. The calibration is not Gaussian-specific. On four public datasets the Laplace frontier $\sigma^*=d/\ln(A/\varepsilon)$ behaves like the Gaussian one. (a) Both tails hold the interference budget label-free (tail/ε<*1*). (b) The Laplace σ^* reaches parity with a labelled grid search anchored to the Laplace scale.

Reaching Attention And Prototype Networks

The certificate uses only that a prediction is a superposition of distance-decaying weights over stored points. That description is not special to density estimation: it fits two of the most widely used readouts in modern deep learning, one of which is not obviously a kernel field at all. We show both inherit the certificate, and we verify it on real trained networks.

The normalised certificate.

The base frontier (Section 1.1) bounds the *unnormalised* tail $\Sigma_{i\in F} v_i K_\sigma(\cdot)$. Softmax attention, prototypical networks and Nadaraya–Watson regression are instead *normalised* Gibbs fields,

$p_i = e^{-E■/}/\Sigma_j\ e^{-E■/}$, where the output-relevant quantity is the normalised far mass $m_F = \Sigma_{i\in F} p_i$; deleting the far set rescales the denominator, so the unnormalised bound does not transfer verbatim. A companion certificate bounds m_F directly: with a near anchor at minimal energy and a far set separated by an energy gap , $m_F \le\ D_q^F()\ e^{-/}$. Since D_q^F is itself temperature-dependent, the executable form mirrors the additive one-shot theorem: measure the Hill number $D_{q,0}$ once at a reference temperature $_0$, and deploying at $_{cert} = min(_0, /ln(D_{q,0}^F/\varepsilon))$ guarantees $m_F \le \varepsilon$ (Theorem 2, SI Note 13; D_q^F is the Rényi/Hill effective mass of the far weights, of which the participation ratio N_{eff} is the $q=2$ member, SI Note 12). This has no source-amplitude factor--it is exactly the unbounded-V_{max} obstruction that made attention the weakest case above, now removed--and for a prototypical network ($E_i = ||\ f(x)-c_i||^2$) and softmax attention ($E_i = -\ q,k_i$) it holds exactly. The two views are complementary: the additive frontier certifies the interference *tail* in output units; the Gibbs certificate certifies the normalised far *mass* that a softmax reads.

Beyond kernel density: a transformer's attention.

The predictors above are all built as kernel fields by construction. Softmax attention is not--its weight on key *i* is $(\ q,k_i/)$, an inner product, not a distance. Yet the polarisation identity $q,k_i = 12(||q||^2 + ||\ k_i||^2 - ||\ q-k_i||^2)$ makes the query-only factor cancel in the softmax, leaving *exactly* a Gaussian kernel field over the keys with bandwidth $\sigma = \sqrt{()}$ and per-source weight $v_i = e^{||\ k■||^2/2}$ (SI Note 9). This algebraic identity is what lets a distance-based certificate reach attention at all; but because the induced weights $v_i() = e^{||\ k■||^2/2}$ themselves vary with , the additive frontier does not transfer as a fixed-temperature deployment guarantee, and the operational attention certificate is the normalised Gibbs bound below (Theorem 2), which is stated directly in the logit gap and is temperature-explicit. As a descriptive reading, the additive frontier still fixes a certified temperature $^* = (\sigma^*)^2 = d^2/(2ln(A/\varepsilon))$ from the field's geometry. We test this on the real query/key vectors of every sampled head of two pretrained transformers--BERT-base and GPT-2--over long contexts (*360* keys per head). The identity is exact to the models' float32 precision ($<10^{-4}$ across all heads), and at * the certified far-key interference tail stays within budget across *35,000* queries per model with *zero* violations, overshooting by *18–33×* when is pushed to 4^* (Fig. 6). Attention's source weights v_i are unbounded, so this is the interference-tail certificate (Proposition 2) rather than the full $V_{max}=1$ locality readout, which transfers cleanly only when key norms are bounded--the pure-Gaussian case (the measured normalised far mass at * is *0.35* for BERT and *0.62* for GPT-2, not ε-small, but well below the *0.80–0.84* at 4^*). Reading attention through a kernel is not itself new--the polarisation view and the linear-attention/Performer line make the same identification ; what is new here is that the label-free interference certificate transfers onto that kernel, giving attention a computable far-key guarantee it did not have before. One caveat on deployment: in a pretrained transformer is fixed at $\sqrt{(d_k)}$, so * is the certified temperature the geometry *implies*, and our BERT/GPT-2 numbers are a descriptive certification of that field, not a claim that the running model was retuned to *. The normalised-mass companion (Theorem 2) certifies the same heads without the unbounded-weight caveat: at the deployed $=\sqrt{(d_k)}=8$ the measured normalised far mass is *0.24–0.27*, and the logit-gap temperature that would drive it below ε is far cooler (median *≈0.02*), quantifying exactly how much more peaked attention would have to be to be provably local--validated with zero violations across *66,000* query-heads on both models (SI Note 13).

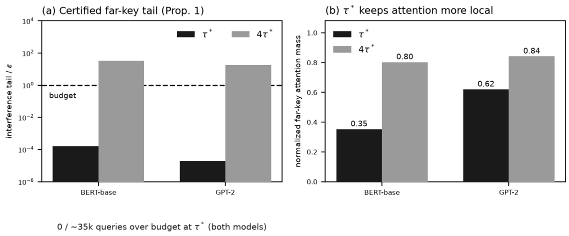

Figure 6. Attention inherits the certificate. On real query/key vectors from BERT-base and GPT-2 (all sampled heads, *360* keys each). (a) At the certified temperature * the far-key interference tail is far below budget (mean $<10^{-3}\varepsilon$; zero of *35,000* queries per model exceed it) and blows up *18–33×* at 4^*. (b) * holds attention markedly more local than an over-wide temperature.

From a single head to the full attention layer.

A single head is not the full attention layer. Multi-head attention runs *H* heads in parallel, mixes each through a value matrix W_V^h, and combines them through an output projection W_O; the certificate must survive the combination of heads and projections--and it does, with a short argument. At * each head's far keys carry normalised mass m^h (the single-head far mass of Note 9); deleting the far set moves that head's output by at most $2m^h||\ W_V^h||_2\ X$ ($X = max_i||\ x_i||$), and stacking through W_O gives $||O||_2 \le ||\ W_O||_2(\Sigma_h(2m^h||\ W_V^h||_2\ X)^2)^{1/2}$ (Prop. 5, SI Note 11)--a layer-output locality bound computable from the pretrained weights and the per-head masses alone, no labels and no gradient. On a real `bert-base-uncased` layer ($H=12$) the bound holds at every sampled query across four layers, with the actual far-deletion perturbation a non-trivial *12–28%* of the layer output and the bound conservative by a measured *50×*. Honestly scoped: because attention's weights are unbounded the far masses are the measured *0.24–0.38*, not ε, so this certifies a *bounded, computable* layer perturbation rather than an ε-small one--a label-free, geometry-only far-field guarantee for the full attention layer, loose by a measured factor but not vacuous.

The bounded-weight companion: prototypical networks.

Attention exercises the unbounded-weight edge of the class; a prototypical network exercises the clean interior. It classifies a query by $softmax_k(-||\ f(x)-c_k||^2/)$ over class prototypes c_k--already a Gaussian kernel field, with *no* polarisation needed and all source weights $v_k=1$, so assumption (A1) holds with $V_{max}=1$ exactly. Both the interference-tail certificate *and* the full far-set locality readout (Corollary 2) therefore transfer. We train a *4*-layer convolutional ProtoNet on Omniglot (test accuracy 91.5%) and certify its prototype fields on held-out *60*-way episodes (*12,000* queries). On the *unnormalised* field ($v_k=1$) the additive certificate applies at full strength: the far-prototype tail sits at *0.25*ε with *zero* violations, and the full far-set locality readout (Corollary 2) transfers because the weights are bounded--the guarantee attention cannot get. The *normalised* softmax probabilities are certified separately by the Gibbs theorem (Theorem 2), which is the correct object for a normalised readout and which we validate on this architecture in SI Note 13. Attention and prototypes thus bracket the two ends of the same class: attention's unbounded weights admit only the interference-tail bound, while the bounded-weight prototype additionally inherits the full locality readout.

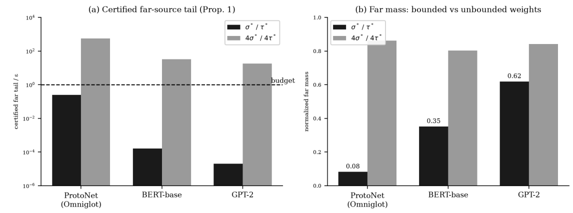

Figure 7. A family of readouts inherits the certificate. Prototypical networks (Omniglot) alongside the two transformers. (a) Every architecture holds the certified far-source tail below budget at its own $\sigma^*/^*$ and overshoots at *4*× (*18–33*× for attention, *560*× for ProtoNet). (b) The normalised far mass separates the bounded case (ProtoNet, $v_k=1$, far mass *0.08*--ε-small) from the unbounded case (attention, *0.35–0.62*): the full locality readout transfers exactly only when weights are bounded.

The Budget Travels: Re-certification Without Labels

The certificate re-certifies a field it has never labelled or tuned on, which a labelled search cannot do once its labels are spent. A labelled search spends its labels at tuning time, on the field it saw; once deployed on a field that has grown--as continually-updated models and growing memories do--it has no labels left to re-tune and no way to know whether its fixed bandwidth still holds the budget. Because σ^* re-measures the mass *A* from the field itself, it re-certifies for free. Re-measurement is what any label-free interference rule can do--a direct tail search would re-certify too (see Related work)--but σ^* carries an a-priori conservativeness proof into each new stage, which the empirical searches do not. This is a capability the labelled-search category lacks rather than a higher score against a baseline.

The claim is a structural gap against a bandwidth frozen after one labelled tuning stage; the numbers measure it. We grow the field by density--from *3* to *120* examples per class (*300* to *12,000* centres)--on all three frozen backbones, fixing a labelled grid-search σ once on the sparse initial field. As the field densifies, that frozen bandwidth becomes an over-smoother whose interference drifts out of budget while its accuracy stalls; σ^* re-measures *A*, shrinks to track the density, and holds tail/ε≈*0.1* throughout (Fig. 8). On ResNet-18 the frozen σ=*8.83* climbs to tail/ε=*502* and 47.1% accuracy while σ^* reaches 54.1%--a *7*-point label-free margin; on ViT-B/16 the frozen tail reaches *3865*× budget and σ^* leads by *6.1* points. ResNet-50 is the honest boundary: its geometry keeps the frozen bandwidth near budget (tail/ε peaks at ≈*2*, a mild crossing not a collapse), yet σ^* still gains *3.2* points. Safety-collapse rests on two backbones, the accuracy gain on all three.

Two guardrails, up front because promotion invites scrutiny. First, this is *adaptivity, not dominance*: the margin is against a bandwidth *deliberately frozen* on the sparse field. Against a per-stage re-tuned oracle σ^* does not win; it *tracks* it (within a fraction of a point on ResNet-18 and ViT, trailing by ≈*4* on ResNet-50), label-free--its advantage is needing no labels to re-tune. Second, the win is *regime- and mode-specific*: it appears in the dense, growing regime (at fixed low density σ^* gives parity, below), and it follows *densification*, not field size. A class-incremental run adding classes at fixed density (*10→100* classes, *120* each) barely moves the geometry (*d* drifts *-3.5%*), so the frozen bandwidth stays in budget and ties σ^*--a clean null we report to fix the boundary. The certificate travels when local density changes, the growing-memory setting; it adds nothing when only the class count grows.

Public datasets.

On iris, wine, breast-cancer, and digits the naive bandwidth overshoots by 2.8–147× while σ^* holds tail/ε<*1*. These fields are neither dense nor growing, so the labelled grid search is a fair fixed comparator, and σ^* matches it within about two points on three of four (breast-cancer 96.6 vs 96.5, digits 97.0 vs 97.0, iris 96.3 vs 96.3%); on wine it trails by 2.2 points. σ^* is a specificity-first bandwidth: at fixed low density it delivers parity, and in the dense, growing regime the guarantee turns into an accuracy win.

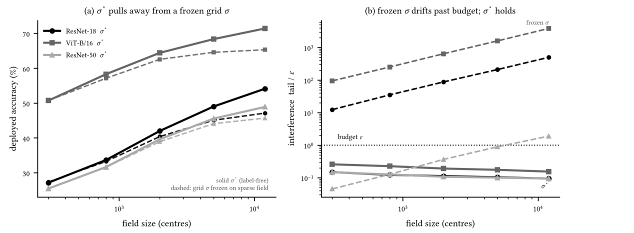

Figure 8. The budget travels--and buys accuracy, on three backbones. A labelled grid-search σ is fixed once on the sparse *3*-per-class field, then the field densifies to *120* per class. Solid: σ^* (label-free, re-measured each stage); dashed: the frozen grid σ. (a) σ^* pulls away from the frozen bandwidth as the field grows (ResNet-18 *+7*, ViT-B/16 *+6.1*, ResNet-50 *+3.2* points at full density), tracking a per-stage re-tuned oracle. (b) The frozen bandwidth's interference drifts past the budget (ResNet-18 *502*×, ViT *3865*×; ResNet-50 a milder crossing at ≈*2*×) while σ^* holds tail ≈*0.1*ε throughout.

Discussion

A globally supported influence system can be made certifiably local by calibrating its operating resolution to the effective mass of its distant sources. For a kernel field that calibration is closed-form, $\sigma^*=d/\sqrt{2\ln(A/\varepsilon)}$, with the mass *A* measured from the field rather than tuned; it needs no labelled search, and it carries an a-priori per-query guarantee--that every prediction it flags stable is decided by its intended sources--that an empirical search does not. Because the mass is measured rather than tuned, the certificate re-certifies a field it has never labelled or tuned on: as the field densifies, the same label-free re-measurement tracks a per-stage re-tuned oracle on three backbones, while a bandwidth frozen on the sparse early field drifts to hundreds of times the budget in the growing-memory regime where labels are no longer available. And because the certificate needs only distance-decaying weights, the same calibration reaches objects never built as kernel fields: softmax attention and prototype networks inherit it--the full locality guarantee for the bounded-weight prototype, the temperature-explicit far-mass bound for attention--placing a label-free locality guarantee on distance-based deep readouts. The bandwidth question is thus a calibration, not a search: derive the certified resolution from geometry, and read off the guarantee. Because that certificate is differentiable and depends only on geometry, a natural next step is to make it a training objective--penalising the far tail, or clamping the bandwidth to σ^*, so a model is locality-certified by construction rather than after the fact; we leave that to future work.

Related Work

Density-estimation rules--plug-in bandwidths (Silverman, Scott), Sheather–Jones , variable-bandwidth and diffusion selectors , Lepski's method , and scale-space --target an integrated squared error against a true *density*; we target *interference*, and derive the bandwidth from that objective using no labelled signal. Run on the three frozen backbones, these objective-mismatched selectors miss the budget in *opposite* directions: Sheather–Jones under-smooths to a near-delta kernel that collapses to a *1*-nearest-neighbour rule (a numerically stable readout recovers its *1*-NN accuracy; the naive linear-domain evaluation underflows to chance, an artifact we avoid), while

Gaussian-process marginal-likelihood over-smooths to $11–15\times\sigma^*$ (tail/$\varepsilon\,3\times10^5$)--confirming that likelihood- and ISE-driven selection is blind to the interference budget in either direction. The label-free distance heuristics--the median heuristic and local scaling --fix the scale but not the safety factor, and overshoot the budget by roughly five orders of magnitude on the dense fields (Table 1). A sharper label-free comparator targets the interference directly: since $Tail_\sigma(y,d)$ is monotone in σ and costs the same kernel evaluations as measuring A, one can *bisect* σ until the measured mean tail equals ε--an empirical search for the same frontier, with no closed form and no proof. We run it on all three backbones. It is not free lunch: because σ^* is provably conservative (its mean tail is only $0.08–0.16$ ε), bisection deploys at a *larger* bandwidth ($1.07–1.11\times\sigma^*$), and on these high-dimensional fields the labelled accuracy optimum lies *below* σ^*, so moving up costs $0.8–2.1$ accuracy points rather than recovering them--bisection is further from the labelled grid than σ^*, not closer. More importantly it forfeits the guarantee: holding the *mean* tail at ε leaves the per-query distribution over budget (p99 tail/ε $1.6–2.3$, max up to 2.8), whereas the mean-A form of σ^* certifies the mean tail with an a-priori conservativeness proof, and its quantile-A form certifies the covered $(1$-$\alpha)$ fraction by construction (the maximum-A form holds every measured query, SI Note 5), with the conformal wrapper (Proposition 7) supplying the finite-sample future-query guarantee. The distinction the paper claims is thus precise: σ^* is a one-shot, closed-form, provably conservative certificate with an a-priori per-query form available; a label-free tail search is an empirical mean-fit with no closed form and, here, worse accuracy. Gaussian-process length scales (learned by the marginal-likelihood optimisation measured above) have motivated uniform error bounds for safety-critical use ; σ^* fixes the scale from geometry and offers an a priori certificate rather than a fitted posterior. Conformal prediction and selective classification certify *marginal* coverage from a *labelled* calibration set, whereas Corollary 1 certifies *per-query decision equality* without labels--both require a representative sample (label-free is not sample-free: A is measured over held-out queries), and the two are complementary, a conformal wrapper calibrating the residual far-field probability our margin leaves uncontrolled. In continual learning , where a correction field is adapted over a frozen model, the interference budget makes the stability–plasticity trade-off quantitative. The locality readout (Section 1.1) relates to machine unlearning and influence-based attribution : those quantify how removing a training point changes the output, while σ^* *certifies*, label-free and a priori, that removing any far point changes the output by at most ε and corrupting it cannot flip a 4ε-margin decision (Corollary 2)--a measured data-independence guarantee for distance-based predictors, distinct from the randomised, worst-case sensitivity of differential privacy.

Which models inherit the certificate.

The certificate uses only that the output is a weighted sum of training points with a distance-decaying weight, so it is a property of a *family* of modern readouts, not of kernel density estimation alone. We demonstrate two members on real networks, chosen to bracket the class. Softmax attention reduces exactly to a Gaussian kernel field over keys with unbounded source weights; its certified far-key tail holds on BERT and GPT-2 (Fig. 6). Prototypical networks are the bounded-weight companion ($v_k=1$), for which the full locality readout transfers, not just the tail; the certificate holds on a trained ProtoNet with an ε-small unnormalised far tail (0.25ε), and its normalised probabilities are certified by the Gibbs theorem (Fig. 7, Theorem 2). Between them they instantiate both ends of the assumption (A1). Two further families remain candidates we do *not* test here: representer-point decompositions write a network's logit as such a field, and a network near convergence acts as a kernel machine in its neural-tangent space . For each, σ^* would supply a candidate locality radius from the field's own geometry; verifying the budget on them is future work.

LIMITATIONS

Three boundaries. On pure density recovery there is no near/far structure--every sample is near--so the separation objective is silent; the principle applies to fields that must *discriminate*. σ^* is specificity-first: its guarantee is on interference and accuracy dominance is regime-dependent (parity at fixed low density, an accuracy win only in the dense, growing regime). The re-certification advantage is likewise mode-specific--it follows density, not field size: a class-incremental run that adds classes at fixed density leaves the geometry, and a frozen bandwidth's budget, essentially unchanged (a null we report in Section 1.4), so the growing-field claim is scoped to densification, the growing-memory regime, not to class count alone. And conservativeness (Proposition 2) is stated at fixed radius d; d drifts by only 1% over a $40\times$ densification, consistent with $O_p(n^{-1/2})$ quantile convergence (SI Note 5), so re-estimating it per stage corrects a vanishing term, but a field whose class-conditional geometry itself changes would need d re-derived.

METHODS

Construction.

For each backbone--ResNet-18 (512-d), ResNet-50 (2048-d), and ViT-B/16 (768-d), all ImageNet-pretrained and frozen--we extract the penultimate features of CIFAR-100 and build a correction field of $20{,}000$ balanced kernels (200 per class) over the training features, classified by a kernel-weighted (Parzen) vote. The locality radius d is the tenth percentile of pairwise centre distances--pure geometry, no labels. We fix $\varepsilon=0.05$ a priori and compare three bandwidths: the naive single-interferer rule $\sigma_{pair}=d/\sqrt{2\ln(1/\varepsilon)}$; the certified frontier σ^*, whose mass $A=V_{max}N_{eff}$ is *measured* once at σ_{pair}; and, as an oracle reference, a labelled grid search over a 30-point log-spaced grid (denoted ``grid"). We also run the classical label-free density bandwidths--Silverman, Scott, and Abramson's square-root law.

Computing σ^* (label-free).

Four steps, no labels: (1) fix ε and set d to the tenth percentile of pairwise centre distances; (2) form $\sigma_{pair}=d/\sqrt{2\ln(1/\varepsilon)}$; (3) measure $A=V_{max}N_{eff}$ at σ_{pair}, averaging the far-mask participation ratio over queries; (4) return $\sigma^*=d/\sqrt{2\ln(A/\varepsilon)}$. By Proposition 2 measuring at $\sigma_{pair}\geq\sigma^*$ over-estimates A, so one pass suffices. Full computational specification, grids, and seeds are in SI Note 7 and the repository.

Reproducibility.

The full computational specification--feature extraction, the standardisation and distance conventions, the bandwidth grids and seeds for every experiment, and the growing-field and modality protocols--is given in Supplementary Note 7, and all scripts and released result files are in the repository. Proofs of every proposition, theorem, and corollary stated above are in the Supplementary Information (Notes 1–14).

Data and code availability.

All experiment scripts and released result files are available at `github.com/negulescu42/sigma-star` with fixed seeds and the exact bandwidth grid for each experiment. Every empirical certificate reported here--including the maximum-tail and zero-exceedance claims--is reproducible end-to-end from these scripts on the stated frozen features.

Author contributions.

R.N. conceived the resolution-calibration framing and developed the empirical and machine-learning program (Sections 1.2–1.4). C.B. is responsible for the analytic core: the safety frontier (Proposition 1), the conservativeness theorem (Proposition 2), the decision-stability theorem (Theorem 1), the far-set locality corollary (Corollary 2), the uniform-over-region certificate (SI Note 4), the Rényi/Hill effective-mass family (Proposition 6, SI Note 12), the normalised Gibbs far-mass certificate (Theorem 2, SI Note 13), and the conformal finite-sample query certificate (Proposition 7, SI Note 14). R.N. carried out the transformer, prototype-network, and re-certification experiments validating these results. Both authors reviewed and approved the final text.

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